

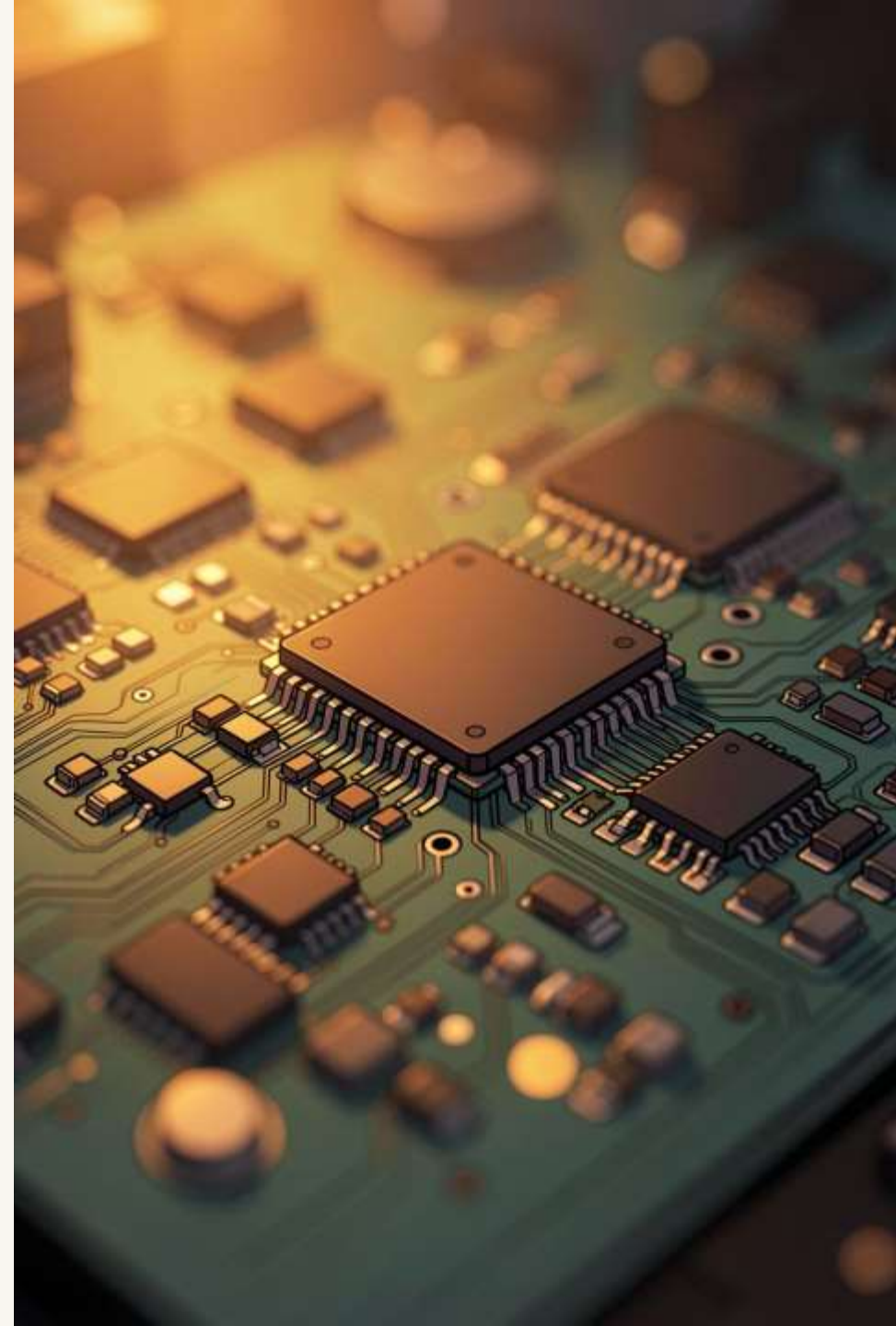
Fuzzy Logic in Artificial Intelligence

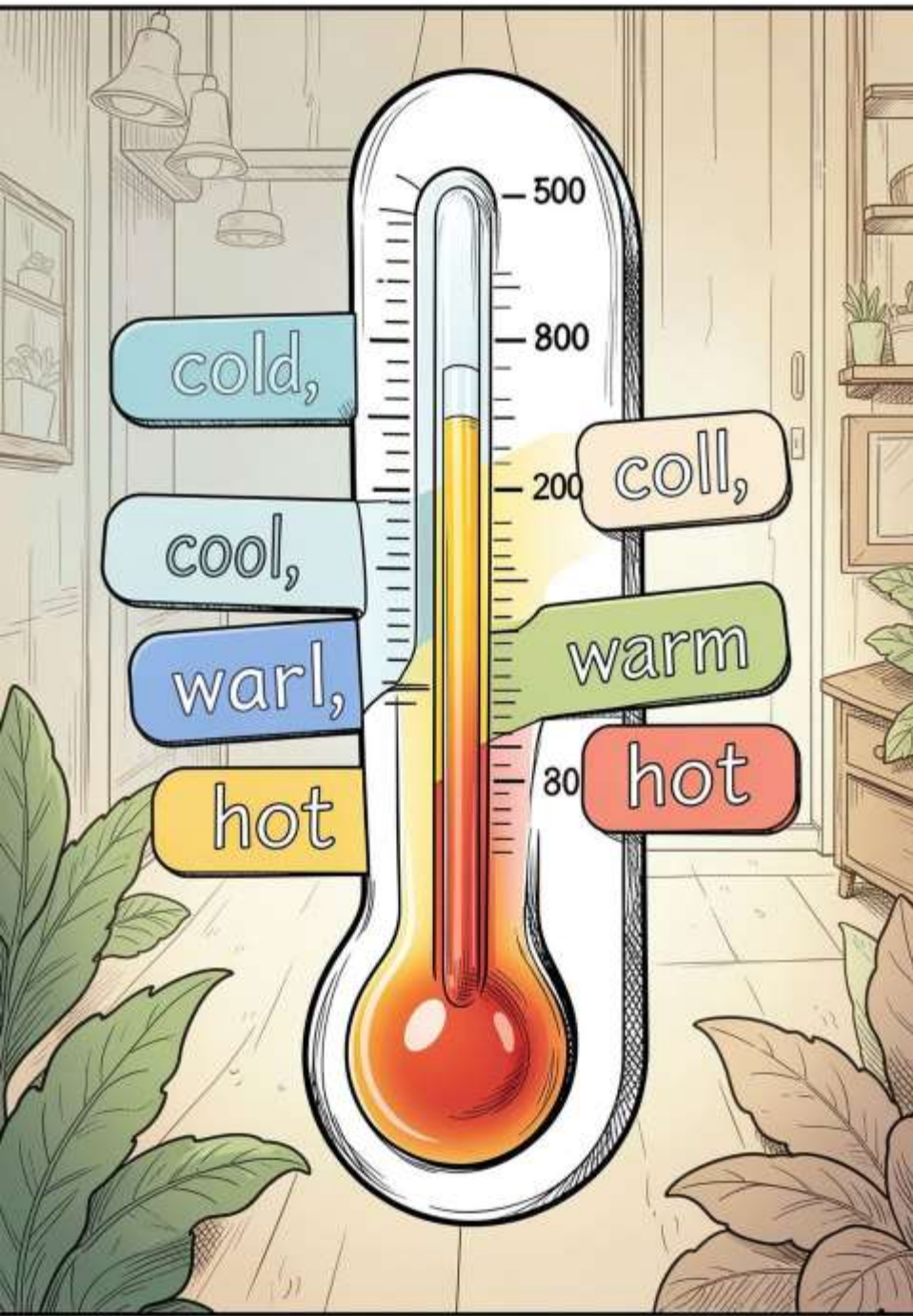
Artificial Intelligence

CSA-1710

Presented by:

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Introduction — What is Fuzzy Logic?

- Fuzzy logic models reasoning that is approximate rather than exact, enabling decisions with unclear or partial information.
- Instead of only True/False, it uses membership values on a continuous scale between 0 and 1.
- Example: 30°C could have membership 0.7 in "Hot" and 0.3 in "Warm".



Useful when binary logic can't capture human-like judgments.

Simple Example — Automatic Fan



Fuzzy logic maps measured temperature to linguistic states and blends them to choose a fan speed.

When temperature = 30°C, the controller treats it as partly warm and partly hot, producing a blended speed rather than an abrupt switch.

Medium / High

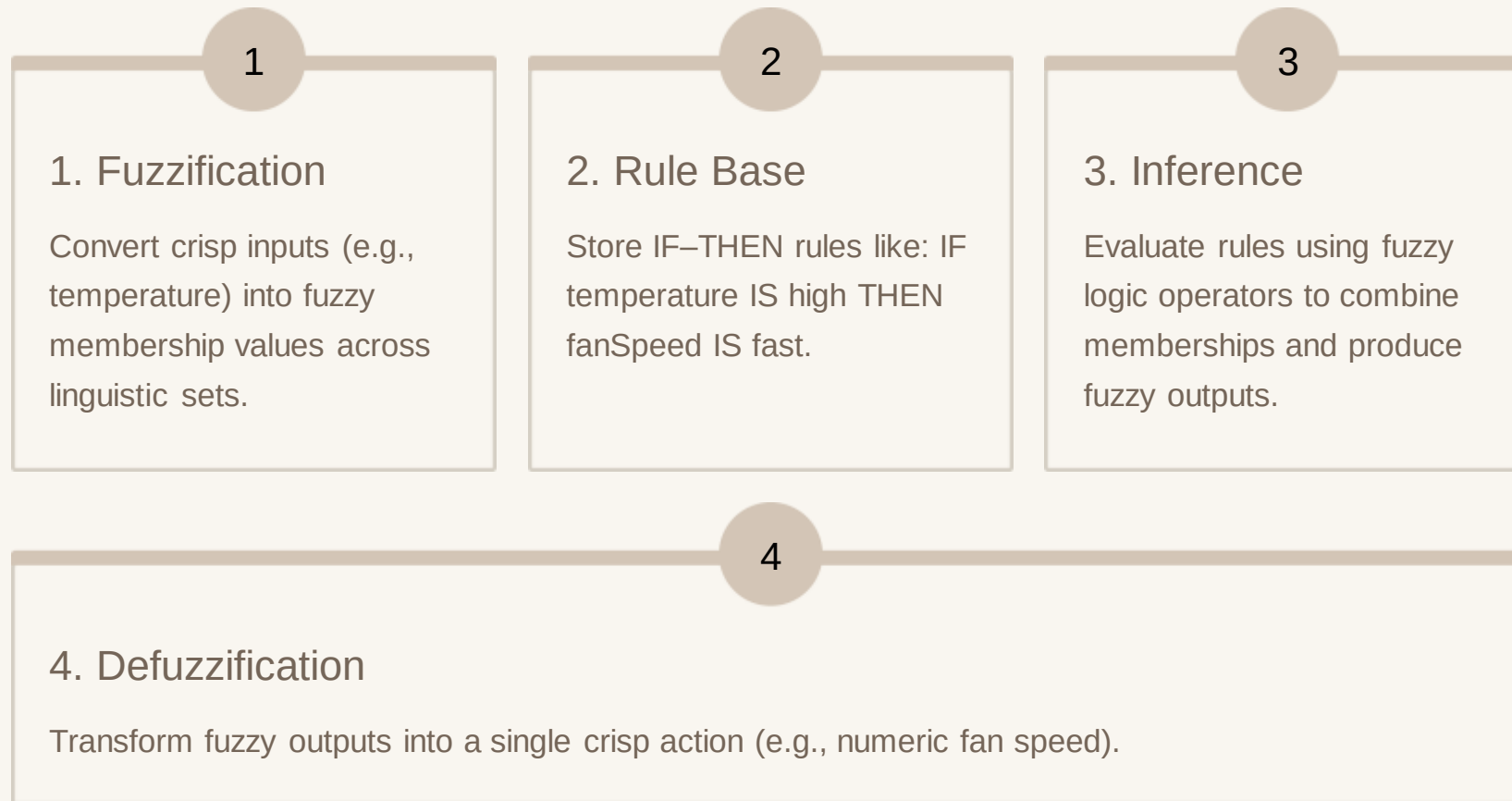
Overlap between warm and hot allows proportional fan speed between medium and high.

Low

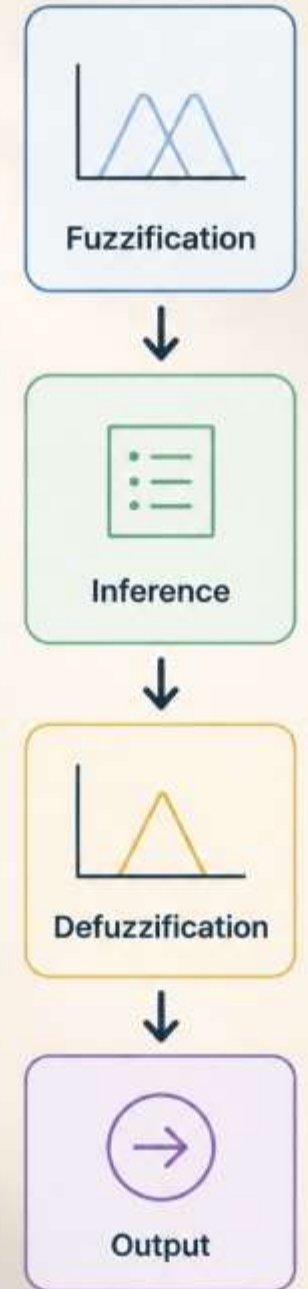
Low temp → low fan speed
(membership strong near cool temps)



How Fuzzy Logic Works — High-Level Process



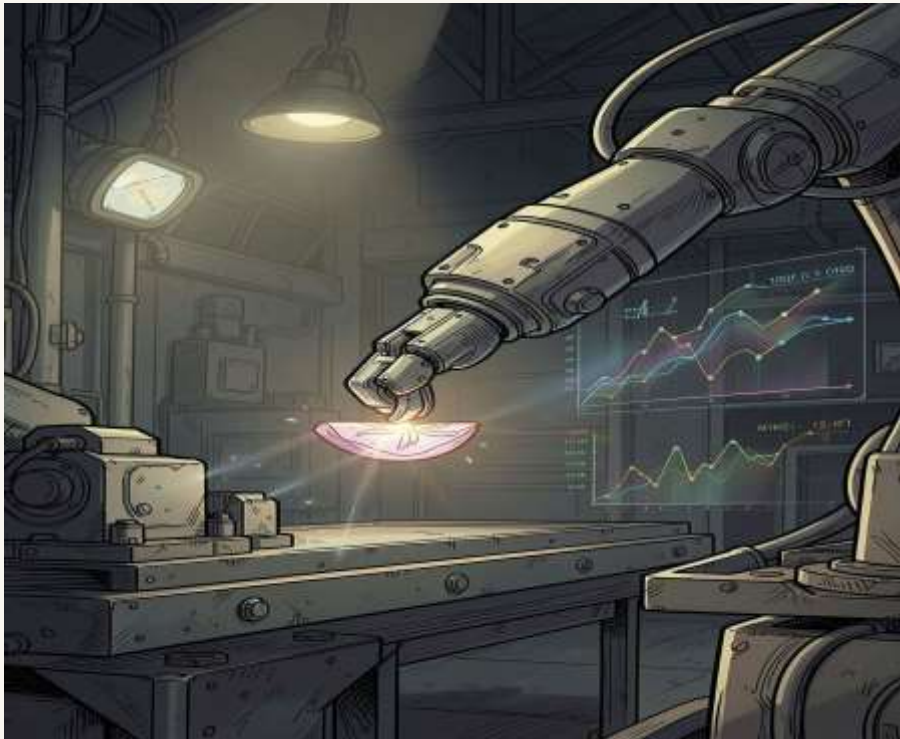
Flow: Input → Fuzzification → Rules → Inference → Defuzzification → Output





Real-World Applications

- Air conditioners — nuanced temperature control for comfort and efficiency.
- Washing machines — adapt wash time and water level to load and soil level.
- Automotive systems — smooth gear selection, stability control, and driver assistance tuning.
- Robotics — graded decision-making in unstructured environments.
- Traffic control — adaptive signal timing responding to flow variability.
- Cameras — autofocus and exposure adjustments using fuzzy criteria.



Fuzzy systems excel where explicit mathematical models are costly or unavailable.

Advantages of Fuzzy Logic

Intuitive

Rules mirror human reasoning and are easy to understand and tune.

Handles Uncertainty

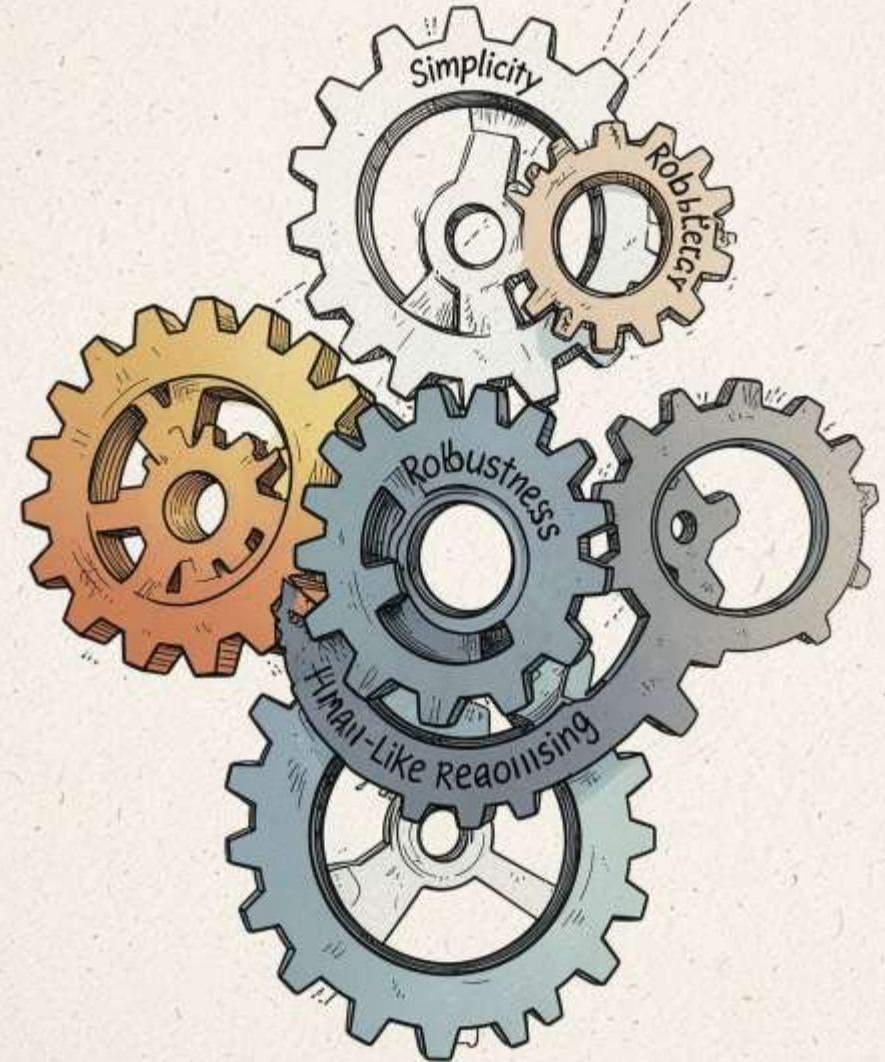
Gracefully deals with vague, noisy, or incomplete inputs without strict thresholds.

Model-Light

Often no precise physical model required; designers encode expert heuristics.

Smooth Control

Produces gradual, stable outputs—avoids chattering from binary logic.



Practical Tips for Building Fuzzy Controllers



- Start with simple membership functions (triangular/trapezoidal) and add complexity only as needed.
- Keep the rule base compact — prioritize high-impact rules and test interactions.
- Use simulation and real-world tuning iteratively to refine membership shapes and defuzzification methods.
- Combine fuzzy layers with conventional algorithms (PID, state machines) for hybrid control.



Good design balances interpretability, performance, and implementation constraints.



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Thank You!

